

DRAFT – IN WORK – NOT APPROVED FOR PRODUCTION USE

STANDARD OPERATING PROCEDURE

Hydraulic Press Setup and Changeover

Die Change and Setup of 250 t Hydraulic Presses

Document number	SOP-00141
Title	Hydraulic Press Setup and Changeover
Revision	A.1 (revision A, iteration 1)
Lifecycle state	In Work
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Document owner	R. Okafor, Production Engineering
Approved by	Pending – in review
Applies to	Greenfield Plant – Press Shop, presses HP-250-01 and HP-250-02

1 Purpose

This procedure describes the setup and die changeover of the 250 t hydraulic presses in the Press Shop. It ensures that dies are installed safely, that tonnage and shut height are set correctly for each die and that the first part is approved before production starts.

NOTE This document is a draft for review. It must not be used for production until it has been released.

2 Scope

- 2.1

This procedure applies to the 250 t hydraulic presses HP-250-01 and HP-250-02 and to all dies in the die library that are released for these presses.
- 2.2

Die maintenance and repair in the tool room are outside the scope of this procedure.

3 References

Document	Title
SOP-00087	Lockout/Tagout (LOTO) Procedure
DSS	Die Setup Sheet (one per die, identified by the die number)
F-141-01	Press Changeover Checklist
F-141-02	First-Article Inspection Report
MM-HP250	OEM Operating Manual, 250 t hydraulic press
ISO 16092-3	Machine tools safety – Presses – Part 3: Hydraulic presses

4 Responsibilities

Press setter: Performs the changeover as an authorized employee under SOP-00087, sets tonnage and shut height from the Die Setup Sheet and performs the safety device checks.

Press operator: Assists the setter as directed and starts production only after release by the setter and Quality.

Quality inspector: Performs the first-article inspection and releases or rejects the setup.

Production Supervisor: Plans changeovers and ensures that only qualified setters perform them.

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5 Safety / PPE

WARNING Crushing hazard. Never place any part of the body between the dies unless the press is locked out in accordance with SOP-00087 and safety blocks are installed between the ram and the bolster.

- 5.1 Die changes shall be performed with the press locked out in accordance with SOP-00087, except for the steps that require power, which are performed in inch mode from the two-hand control station.
- 5.2 Safety blocks shall be installed whenever work is performed between the dies.
- 5.3 Dies shall be lifted only with the 2 t overhead crane and certified lifting equipment; nobody may stand under a suspended load.
- 5.4 Safety glasses, safety shoes (S3), cut-resistant gloves for handling dies and blanks, and hearing protection must be worn in the Press Shop.
- 5.5 The light curtain and the two-hand control shall never be bypassed.

6 Tools & Materials

Item	Specification
Overhead crane	2 t, with certified slings and die lifting eyes
Die cart	Rated 2 t, with locking casters
Safety blocks	Rated for the press, yellow, with interlock plug
Torque wrench	50–350 N·m, within calibration date
Light curtain test rod	30 mm diameter (light curtain resolution)
Shut height gauge	Digital, 0.01 mm resolution

7 Procedure

7.1 Preparation

- 7.1.1 Confirm that the die number on the production order matches the die and the Die Setup Sheet (DSS).
- 7.1.2 Check that the die carries a green "Ready for Use" tag from the tool room.
- 7.1.3 Read the required tonnage, shut height and cushion pressure from the DSS and enter them on F-141-01.
- 7.1.4 Target changeover time: TBD (to be agreed with Production).

7.2 Removal of the previous die

- 7.2.1 Lower the ram in inch mode until the upper die rests on the lower die.
- 7.2.2 Release the upper die clamps and raise the ram to the top position.
- 7.2.3 Lock out the press in accordance with SOP-00087 and install the safety blocks.
- 7.2.4 Release the lower die clamps, move the die onto the die cart and return it to the die library.

7.3 Die installation

- 7.3.1 Clean the bolster and the ram face; remove slugs and debris.
- 7.3.2 Place the new die on the bolster with the die cart and locate it with the locating keys.
- 7.3.3 Clamp the lower die. Tighten the M16 clamp bolts to 210 N·m in a cross pattern.
- 7.3.4 Remove the safety blocks and the locks in accordance with SOP-00087.

7.4 Shut height and tonnage

- 7.4.1 In inch mode, lower the ram onto the upper die, clamp the upper die to the ram and tighten the clamp bolts to 210 N·m.

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- 7.4.2** Set the shut height to the value on the DSS. The shut height must be within the press range of 450–750 mm and within ± 0.1 mm of the DSS value.
- 7.4.3** Set the tonnage limit to the value on the DSS. The tonnage setting shall not exceed 80 % of the rated press capacity (200 t on a 250 t press) unless Production Engineering has approved a higher value in writing.
- 7.4.4** Set the die cushion pressure to the value on the DSS.

7.5 Safety device checks

- 7.5.1** Two-hand control: Verify that the ram does not move when only one button is pressed, that it moves only when both buttons are pressed within 0.5 s of each other, and that releasing either button stops the ram.
- 7.5.2** Light curtain: With the ram moving down in single-stroke mode, pass the 30 mm test rod through the protective field at the top, the middle and the bottom of the field. The ram must stop each time.
- 7.5.3** If any safety device check fails, the press shall be tagged "Do Not Operate" and the Production Supervisor and Maintenance informed. Production must not start.
- 7.5.4** [Review note: confirm the minimum safety distance of the light curtain with EHS before release.]

7.6 Trial strokes and first-article inspection

- 7.6.1** Make two trial strokes in single-stroke mode and check the parts visually for splits, wrinkles and burrs.
- 7.6.2** Submit the first part produced at nominal settings to the Quality inspector together with F-141-01.
- 7.6.3** The Quality inspector measures the part in accordance with the control plan and records the results on F-141-02. Production may start only after the first article has been approved.
- 7.6.4** The approved first article shall be tagged and kept at the press until the end of the shift.

8 Inspection & Acceptance Criteria

Check	Acceptance criterion
Die identification	Die, DSS and production order match
Tonnage setting	Per DSS; max. 80 % of rated capacity (200 t)
Shut height	DSS value ± 0.1 mm; within 450–750 mm
Clamp bolt torque	210 N·m (M16)
Two-hand control	Ram stops when either button is released
Light curtain	Ram stops at all three test positions
First article	Conforms to the control plan; approved on F-141-02

9 Records

- 9.1** The completed Press Changeover Checklist F-141-01 and the First-Article Inspection Report F-141-02 are filed with the production order.
- 9.2** Records shall be retained for 5 years.

10 Revision History

Rev.	Date	Author	Description of change
A.1	2026-09-02	R. Okafor	Initial draft for review.